

Test Manual

Boeing Human-robot interaction for physiotherapy

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Instructions for Filling Out the Template

This test manual describes a collection of tests for the STARS system. This can be the test of a class, a subsystem, a collection of subsystems, or the full STARS system. The description is broken down into the specifications for each test, descriptions of the test, and analysis of the test results. If automated test tools (this includes rules in make files) or special techniques are to be used, they and the conditions for their use are described as well.

Each team needs to describe 4 ± 2 test cases. Each test case must contain or refer to the following components:

- Test Specification
- Test Description
- Test Analysis Report
- Testing Schedule
- Test Materials

This document template assumes the description of 4 tests: Test A, Test B, Test C, and Test D.

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Objectives

This document specifies the acceptance test manual for the STARS system (Group 21's human-robot interaction project for Boeing, applied to a physiotherapy setting). It covers the full pipeline from camera-based person and gesture perception (MediaPipe) through kinematic retargeting (Pinocchio/GMR) to control of the Unitree G1 humanoid robot. Four tests are defined below, each tracing directly to one of the three client-specified acceptance criteria: leader identification and constant-distance following, gesture recognition and reaction, and action mirroring, plus one robustness test for multi-person scenes. Unit tests are run on recorded video and pose data first, integration tests in the Mujoco simulator, and system tests on the physical G1 last, consistent with the client's requirement that all behaviour be simulator-validated before hardware use.

Document References

This manual should be read alongside the Sprint 1 deliverables: Scope of Work, User Stories, Risk Register, and Skills and Resources Audit, which define the requirements and dependencies these tests verify. The acceptance criteria tested here were agreed with the client during the Week 3 client meeting (05/08/2026) and are recorded in that meeting's minutes. The Role Schedule identifies who is responsible for each subsystem referenced in the Testing Strategy and Testing Schedule sections below.

Test Summary

The system under test drives a Unitree G1 humanoid robot so that it can follow a human "leader" at a constant distance, recognise seven defined gestures (turn around, forward, backward, turn left, turn right, go left, go right) and react to them, and continuously mirror the leader's body movements. Three subsystems combine to deliver this: person identification and tracking (MediaPipe pose/skeleton

extraction), gesture and pose interpretation, and kinematic control (Pinocchio/GMR to Unitree SDK). The tests in this document exercise each of these functions individually and in combination.

Testing Strategy

Three subsystems are tested: Person Identification (leader detection and tracking), Gesture Recognition (gesture classification and reaction mapping), and Motion Control (kinematic retargeting and mirroring). Each is unit-tested independently by its owning subgroup, Person Identification by Dhava, Lithasa and Dylan, Gesture Recognition by Christo, Terry and Murphy, on pre-recorded video and motion-capture data, in the team's Ubuntu 22 development environment. Integration testing then runs all subsystems together against the Mujoco simulator, and system testing runs the same scripted scenarios on the physical G1 in a controlled lab space, with the whole team present. Tests are conducted in that order, unit, then simulation, then hardware, so that no untested behaviour reaches the physical robot.

Test A

Test A verifies that the system can identify a single leader in a camera feed containing multiple people, and that the robot maintains a constant following distance from that leader as they move.

Test Specification

Demonstrates the requirement that "the robot should be able to identify its leader and follow them at a constant distance." Tests the person-detection stage of the Person Identification subsystem, the leader-selection rule, frame-to-frame tracking, and the distance-hold behaviour of the motion control stage. Condition: indoor lab space, one designated leader plus zero to two bystanders in frame, run first against recorded video, then in Mujoco simulation, then on the physical G1.

Test Description

Described here for the recorded-video run; the same procedure repeats in Mujoco simulation and then on the physical G1.

A test description consists of:

- **Location of test:** UWA lab space used for team testing; Mujoco simulation environment for the simulated run.

- **Means of Control:** Automatic: leader and bystanders move through frame on foot, no manual test driver.
- **Data:**
 - **Input Data:** Live or recorded camera feed of the leader walking a predefined path (forward, stop, backward, an arc), with 0-2 bystanders crossing the scene.
 - **Input Commands:** None; leader identification and following are autonomous once the test scenario begins.
 - **Output Data:** Per-frame leader ID, bounding box, and measured leader-to-robot distance, logged for the duration of the run.
 - **System Messages:** Console output of the form "Frame N: leader_id=<id>, distance=<value>m".
- **Procedures:** Leader walks a fixed path while the robot (or simulated robot) attempts to maintain a target following distance, defined tolerance $\pm 0.3\text{m}$; distance is logged continuously and plotted after the run.

Test Analysis Report

Not yet run. This section is completed once Test A executes in Sprint 2.

- **Function:** Person Identification correctly designates and tracks a single leader across the test video.
- **Performance:** Distance-hold accuracy: percentage of frames within the $\pm 0.3\text{m}$ tolerance band. Target: 90% or higher.
- **Data measures:** Pending Sprint 2 execution; checked against the 90% distance-tolerance target above.
- **Errors/Deficiencies:** Not yet available; any drift, lag, or leader mis-identification observed during the run will be recorded here.

Test B

Test B verifies that the system correctly classifies each of the seven defined gestures and that the robot reacts with the correct corresponding motion.

Test Specification

Demonstrates the requirement that "the robot should recognise its leader's gestures and react to them (turn around, backward, forward, turn left, turn right, go left, go right)." Tests the Gesture Recognition subsystem's classification accuracy and the mapping from classified gesture to robot reaction command. Condition: leader performs each gesture clearly in frame, one at a time.

Test Description

Described here for the recorded-video run, repeated in simulation and then on hardware, matching Test A's staging.

A test description consists of:

- **Location of test:** UWA lab space; Mujoco simulation for the simulated run.
- **Means of Control:** Automatic: leader performs gestures on camera, no manual test driver.
- **Data:**
 - **Input Data:** Live or recorded video of the leader performing each of the 7 gestures, 3 repetitions each (21 trials total).
 - **Input Commands:** None; gestures are the input, there is no separate command channel.
 - **Output Data:** Classified gesture label per trial and the robot's resulting motion command.
 - **System Messages:** Console output of the form "Frame N: <gesture>", per the client's specified format.
- **Procedures:** Leader performs each of the 7 gestures 3 times in sequence, alternating hands; classification accuracy and reaction correctness are recorded as two separate outcomes per trial, since a gesture can be classified correctly but mapped to the wrong reaction.

Test Analysis Report

Not yet run. Results are added once Test B executes.

- **Function:** Gesture Recognition classifies all 7 gesture types and triggers the corresponding robot reaction.
- **Performance:** Per-gesture classification accuracy and reaction accuracy across the 21 trials; target 90% or higher for each.
- **Data measures:** Pending Sprint 2 execution; checked against the 90% accuracy target above.

- **Errors/Deficiencies:** Not yet available; any misclassifications or incorrect reactions observed during the run will be recorded here, including false positives where a reaction triggers with no gesture given.

Test C

Test C verifies that the robot continuously mirrors the leader's body movements in real time, as distinct from the discrete gesture reactions covered in Test B.

Test Specification

Demonstrates the requirement that "the robot should be able to mirror its leader's actions." Tests the full pipeline from skeleton extraction through GMR/Pinocchio kinematic retargeting to joint commands sent via the Unitree SDK. Condition: leader performs a continuous sequence of arm and body movements, ideally with motion-capture suit data available as ground truth for comparison.

Test Description

Described here for the simulation run first; the physical-hardware run only proceeds once simulation results are acceptable.

A test description consists of:

- **Location of test:** UWA lab space, with motion-capture suit if available for ground-truth comparison; Mujoco simulation for the simulated run.
- **Means of Control:** Automatic: leader performs a fixed 60-second movement sequence (raise arm, bend, turn torso, extend arms) on camera or via the motion-capture suit.
- **Data:**
 - **Input Data:** Camera skeleton stream and, where available, motion-capture suit stream routed through GMR.
 - **Input Commands:** None; mirroring runs continuously from the pose stream.
 - **Output Data:** Robot joint angles over time, recorded for comparison against the leader's skeleton trajectory.
 - **System Messages:** Console log of retargeted joint targets per frame.

- **Procedures:** Leader performs the fixed movement sequence; robot joint trajectories are compared against the leader's skeleton trajectory, first in Mujoco simulation, then, only once simulation results are acceptable, on the physical G1.

Test Analysis Report

Not yet run. This section is completed after Test C executes.

- **Function:** Continuous pose-to-pose retargeting from leader skeleton to robot joint commands.
- **Performance:** Joint-angle error and lag between leader movement and robot response, measured against the recorded skeleton trajectory.
- **Data measures:** Pending Sprint 2 execution.
- **Errors/Deficiencies:** Not yet available; any discrepancies between simulated and real-hardware mirroring behaviour will be recorded here.

Test D

Test D verifies that leader identification remains stable when other people are present in frame, including when the leader is briefly occluded or crosses paths with a bystander.

Test Specification

This is not one of the three headline client requirements but is implied by "identify its leader" in a realistic, non-empty room, and is the most likely failure mode for the Person Identification subsystem. Tests the leader-tracking and re-identification logic under occlusion and crossing scenarios.

Test Description

Described here for the three occlusion and crossing scenarios; each is run against recorded video before any simulation or hardware pass.

A test description consists of:

- **Location of test:** UWA lab space, with 2-3 additional non-leader people present.
- **Means of Control:** Automatic: leader and bystanders move through frame according to three fixed scenarios.
- **Data:**

- **Input Data:** Recorded video covering three scenarios: (1) a bystander crosses in front of the leader, (2) the leader briefly leaves and re-enters frame, (3) two people momentarily overlap in frame.
- **Input Commands:** None; scenarios run as scripted walk-throughs.
- **Output Data:** Leader ID log across each scenario, showing whether the same ID is retained through the occlusion event.
- **System Messages:** Console output flagging any leader ID change or loss of tracking.
- **Procedures:** Each of the three scenarios is run 3 times; leader ID is logged before, during, and after the occlusion or crossing event, and checked for consistency.

Test Analysis Report

Not yet run. Results are added once Test D executes.

- **Function:** Leader ID retention across occlusion and crossing events.
- **Performance:** Percentage of trials where leader ID is correctly retained; recovery time (in frames) after occlusion when it is not.
- **Data measures:** Pending Sprint 2 execution.
- **Errors/Deficiencies:** Not yet available; any incorrect leader hand-offs to a bystander will be recorded here as the most serious failure mode.

Testing Schedule

Unit testing of each subsystem against recorded video and motion-capture data runs through Weeks 3-6, conducted independently by the Person Identification team (Dhava, Lithasa, Dylan) and the Gesture Recognition team (Christo, Terry, Murphy), per the Role Schedule. Integration testing of Tests A-D against the Mujoco simulator runs in Weeks 7-9. System testing on the physical G1 runs in Weeks 10-11, with the whole team present in the lab. A final acceptance demonstration for the client is scheduled for Week 12.

Testing Materials

Unitree G1 humanoid robot; motion-capture suit (Xsens); Ubuntu 22 development VMs and a clean Ubuntu 22 install for final validation; Unitree Mujoco for simulation; MediaPipe, Pinocchio, and GMR repositories; Unitree SDK (Python); a library of recorded test videos covering single-leader, multi-person, occlusion, and re-entry scenarios; a marked floor grid or tape measure for calibrating distance measurements in Test A; VS Code and Git for development and version control.